

5. Venture capital and government policy

Gordon Murray, University of Exeter, UK

INTRODUCTION

“It is instructive to observe that all venture capital markets of which we are aware were initiated with government support. These markets do not appear to emerge without some form of assistance. This leads to the question as to what it is that requires the need for government support in these markets, at least in their formative stages.”

(Lerner, Moore and Shepherd, 2005)

The above quote is taken from a contemporary evaluation of public venture capital activity in New Zealand. It is one of a number of formal reviews of early-stage venture capital activity that have recently been concluded by government policy makers with the assistance of academic researchers¹. The authors of the New Zealand evaluation suggest that, despite venture capital being a financing instrument most widely associated with the “animal spirits” of the free market and of entrepreneurial agents unrestricted by public interference, the state may well have an important role in both initiating risk capital programs as well as providing a conducive environment for the seeding and commercial growth of such activity.

This view of the importance of government commitment to entrepreneurial action, particularly in nascent (usually new knowledge or new technology-derived²) industries, has support from several academic researchers (Botazzi and Da Rin, 2002; Lerner, 2002; Gilbert et al., 2004; Page West III and Bamford, 2005) who see evidence of a significant increase of public initiated and financed venture activity on an international scale. Venture capital and the role of public actors may be seen as one part of such a wider movement to support new enterprise. These authors suggest that the logic behind this

growth of activity is a widening appreciation of entrepreneurship policies “as one of the most essential instruments from economic growth ... for a global and knowledge-based economy” (Gilbert et al., 2004, p. 321). More tangible roles for public intervention are given by among others Lerner, 1999; Jeng and Wells, 2000; Keuschnigg and Nielsen, 2001; 2002; Keuschnigg, 2003. They emphasize the importance of government in setting the supportive legal and economic framework conditions necessary for risk capital activity to flourish. Similarly, Audretsch and Keilbach (2004) sees the entrepreneurial catalyst as the “missing link” in endogenous economic growth theory. Entrepreneurs become the critical conduit for knowledge spill-overs and the subsequent creation of valuable new products and services.

However, academic support for a public role(s) in developing early-stage venture capital markets is, at best, conditional and cautionary (Lerner, 1998). Successful policy makers will have to act with a deft hand. There is plentiful evidence that governments are at least as likely to produce overall negative effects by their involvement in markets as they are to engineer a lasting improvement in market conditions (Gilson, 2003; Armour and Cummings, 2006). Economists are particularly circumspect regarding micro-policy interventions at the “black box” level of the firm or venture capital fund. Their preferred prescriptions are more anonymously concerned with removing market barriers that impede individual agents (e.g. business angels, venture capital firms or entrepreneurs) from pursuing their own commercial interests. But the involvement of government in risk capital markets presumes some form of serious and persistent market failure. Determining the importance or even existence of market failures will, in turn, require a view of both demand and supply side efficiencies. The difficulty of determining market failure, despite its widespread presumption in entrepreneurship finance policy initiatives, is noted below.

Academic and policy interests closely reflect the rapidly growing importance of venture capital activity at both national and international levels over the last 25 years. Governments’ recent endorsement of venture capital’s status as an important instrument of entrepreneurial and innovation policy has been particularly noteworthy in Europe (EC, 1998, 2001, 2003a and b, 2005b, 2006a and b; Murray, 1998; Martin et al., 2002)³. A profusion of contemporary public schemes is also indicative of governments’ contingent

response to the rapid reduction in supply of early-stage risk capital post the Year 2000 collapse in technology markets (Sohl, 2003). High potential young firms were among the first casualties of the changing market conditions at the start of the present century. In several European markets, publicly supported funds were quickly to become one of the most important and continuing sources of risk capital for new enterprises in the hiatus of privately-funded, institutional and informal venture finance following the dot.com collapse (Auerswald and Branscomb, 2003; EC, 2004; NEFI, 2005; Small Business Service/Almeida Capital, 2005).

It would be incorrect to assume that venture capital is exclusively Anglo-Saxon in its nature or distribution although a strong association exists between countries in the former British Empire and the international centers of venture capital activity. Nascent or growing venture capital industries now exist in virtually all developed economies in the world. They are frequently encouraged by government action. Similarly, policy makers in the emerging economies of China, Russia, and Brazil are also now exploring the development potential of risk capital⁴. India has already an established venture capital community⁵.

Despite the assumption that venture capital is a suitable subject for policy action, surveys of SME finance repeatedly show that entrepreneurs' receipt of risk capital from professional investors is an extremely rare event. Reynolds et al. (2003) "guesstimate" that less than half a percent of all nascent entrepreneurs receive either venture capital or business angel finance at start-up. A 2004 survey of UK SMEs showed that less than 2 per cent of respondents had ever raised institutional venture capital (Small Business Service, 2005). A similar percentage has been recorded in the Europe (see EC Flash Barometer, 2005). Given that the UK has the largest and most advanced venture capital/private equity industry in Europe, it is probable that other countries are unlikely to register significantly greater risk capital activity among their young firms. European studies confirm this reality of the scarcity of venture capital receipt (European Venture Capital Association, 2005).

Early-stage (aka "classic") venture capitalists primarily target technology-based young firms because of their potential for very rapid growth in attractive and immature markets. Yet, in a 1997 survey of 600 high tech start-ups in Germany and the UK (i.e.

conducted at the start of the technology bull market), Bürgel et al. (2004) found that only around one in ten of UK firms had received venture capital. In Germany, this ratio went down to one in fourteen. Even in the US, Auerswald and Branscomb (2003) note that the supply of institutional venture capital finance for technology development trails significantly behind business angels, corporations and the Federal Government. It is salutary to note that in 2005, the US and UK venture capital industries invested collectively in only 412 seed and start-ups deals (BVCA, 2006⁶; PricewaterhouseCoopers/National Venture Capital Association, 2006). This is from two major world economies where, collectively, over one million new businesses are started every year. Thus, institutional venture capital still remains a specialist financing instrument of relevance only to a tiny percentage of the population of new and growing enterprises in any economy. This continues to be the case even during a bullish, new technology market.

This chapter will seek to summarize what consensus may be found in seeking an appropriate role and mode of action for government in the light of the evidence of both academic enquiry and policy experience. The question of why governments appear to be so interested in venture capital will be addressed while also noting the considerable influence of US experience. The circumstances under which government itself becomes involved in either influencing or participating directly in risk capital investment will be explained. The nature of policy instruments at government's disposal are subsequently catalogued including the growing interest in "equity-enhancement" programs that incentivise private venture capital agents. The chapter will also seek to address why governments have also become involved in the 'alternative' policy direction of supporting informal investors or business angels. The chapter will conclude by suggesting future research questions of both academic and policy import.

WHY ARE GOVERNMENTS SO INTERESTED IN VENTURE CAPITAL?

The attraction of an established venture capital industry lies in its putative ability to both help finance the creation of new industries and, in so doing, to transform and reinvigorate

mature and established economies (Apax Partners, 2006; European Commission, 2006a and b). The joint application of risk capital and high levels of managerial and entrepreneurial experience is seen as a particularly attractive resource combination (Sapienza, 1992) which possibly explains venture capitalists' popularly perceived ability to both identify and nurture exceptional new and innovative enterprises.

It is evident that the venture capital experience of the United States in the second half for the 20th Century has exerted a huge influence on the entrepreneurship policy ambitions of the majority of developed and emerging economies alike. Above all, it was America's unique ability to generate a stream of new companies of enormous vigor and global span from the nation's advanced science and technology research centers. American venture capital clusters, pre-eminently Silicon Valley and Route 128, are perceived as the "gold standard" of early-stage innovation finance systems (Bygrave and Timmons, 1992). Venture capital – both in its institutional and informal variants – is seen as part of the very fabric of the USA's ability to remain at the forefront of knowledge production and commercialization (Edwards, 1999) through risk capital's contribution to the entrepreneurship/economic growth link (Audretsch, 2004). Multi-country findings from the Global Entrepreneurship Monitor (GEM) further validate venture capital "as playing a central role in facilitating high growth entrepreneurship" (Reynolds et al., 2000).

Accordingly, policy goals are often crudely framed as "How does country X or region Y create the necessary conditions to replicate a Silicon Valley" (Armour and Cumming, 2006). As the emerging BIC economies⁷ grow their global share of the production of manufactured goods of increasing sophistication, high value, knowledge-based goods and services are seen to be of growing importance for the continued prosperity of developed economies. Thus, the question of how to emulate world-class examples of innovative and entrepreneurial excellence remains an urgent policy goal for mature Western economies as traditional markets erode (Archibugi and Iammarino, 1999). In Europe, these concerns about regional competitiveness are exemplified by the Lisbon Agenda which sought by 2010 to make Europe the most competitive world region in which to establish and grow a new business (EC, 2004; Kok, 2004).

Despite the absence of a resolution of the question of how to create a new Palo Alto in Bavaria or a Route 128 around Helsinki, governments' concerns for the continued support of the domestic science base (including the commercialization of intellectual property from laboratory to successful enterprise) remain intertwined with a strong faith in the value of venture capital finance. Venture capital is as much an instrument of innovation policy as enterprise policy. That many governments recognize the importance of venture capital finance is, of course, no argument that they should directly engage in such commercial actions. Most free-market oriented, public administrations have considerable reservations about direct state involvement in specialist financing activities. They would prefer to remove themselves completely from this commercial role. None the less, governments reserve the right to intervene if there is clear evidence that: 1) the supply of appropriate finance is insufficient; that 2) as a consequence, material economic and other benefits from entrepreneurial actions are being lost to the domestic economy; and 3) no private investors will independently increase the supply of risk capital. Thus, public intervention is predicated on an "insufficient" response from private capital markets.

Market failure

We have argued that, given the evolution of the venture capital industry, the competencies and dynamism of its professional managers and the weight of institutional money now at their disposal, there appears little direct role for the state. Yet, there exists a conundrum. As the size and scale of venture capital activity has grown internationally, governments have perversely become increasingly drawn into the investment process. The state has become both a provider of public funds to private venture capital firms and, on occasions, as an active investor directly selecting new enterprises. Governments have by default been obliged to assume responsibilities for early-stage enterprise financing activities that many academic and industry observers believe should better be left to efficient capital markets. Their involvement in the investment process is recognition that the institutional venture capital industry increasingly believes that early stage investments are not sufficiently attractive.

A market failure can be said to have occurred when the price mechanism fails to produce a socially optimal outcome. In effect, rent seeking investors⁸ being unable to capture the full economic and social value of their investments provide less finance (venture capital) than could effectively and profitably be employed in existing opportunities. In these circumstances, public intervention is one possible contingent response to private market shortcomings (European Commission, 2001). At best, the public involvement is seen as temporary and should be designed to be phased out as the venture capital market corrects (OECD, 2004). Yet, the term “market failure” is ambiguous at best. It is frequently used, and mis-used, to argue the case for public intervention and state subsidy in areas where the market appears to deliver less of a product or service than may be deemed desirable by interested parties. Within the context of SME finance, there are repeated calls from entrepreneurs, small enterprise owners and their lobbyists for the state to intervene in order to encourage the providers of finance (typically banks or venture capital firms) to lend more debt or invest more equity capital. These arguments frequently resort to some vague definition of “public good” and are often linked, in the case of venture capital, with arguments promoting internationally competitive, innovation and technology policies. Rarely do such submissions acknowledge that a reduction in supply of finance may be an efficient market’s reaction to an insufficient supply of attractive companies.

Thus, the term market failure is often used as a label rather than a serious argument. The specific market(s) in which the problem occurs needs to be carefully defined. Repeated surveys of finance for SMEs are ambiguous in their findings. For example, a large proportion of small business owners do not require external finance (Small Business Service, 2005). A Eurobarometer survey found that over three-quarters of SMEs has sufficient financing and only 14 per cent of respondents put easier access to finance as their primary concern (EC, 2005). However, it is the much smaller population of high potential and rapidly growing young firms that are most likely to seek external finance. These, immature firms with, as yet, limited assets are also one group that is most likely to find finance is problematic both in its supply and in the cost of access. This is particularly the case for knowledge-based young firms with intellectual and experiential assets that are largely intangible and tacit. (See, for example Bank of England, 1996; Storey and

Tether, 1996; OECD, 1997; Westhead and Storey, 1997; European Commission, 2003a; 2003b; Maula and Murray, 2003 and 2007).

For policy makers, the quandary exists in determining when a constraint in the supply of finance to a potential user is either: 1) an adverse outcome of an inefficient and/or ill informed market; or 2) a rational and well informed judgment by an efficient market on a unattractively priced proposal. In the former case, often called “the equity gap”, there may be an argument for publicly incentivising either economic principals or agents to provide a greater supply of debt or equity (Keuschnigg, 2003). In the latter case, the failure resides in the entrepreneur’s inability to demonstrate the attractiveness of the business proposal. In contemporary policy vocabulary, this venture is not yet “investment ready” (Mason and Harrison, 2001). Here, the prescription is much more likely to be public interventions to improve human capital. A “supply-side” response that seeks to manipulate investors’ returns would not address the core “demand-side” problem of poor quality enterprises

The longevity of the “equity gap”

The term “equity gap” has entered the policy vocabulary. It was first used in an official British governmental report in 1931 that looked at the availability and access of small & medium sized businesses to sources of external finance. The Macmillan Report concluded that firms were facing impediments in the search for capital that were not a function of their attractiveness as individual investment opportunities. Rather, because of their size and designation as small business, owners were facing discriminatory actions by the institutional providers of business finance. Thus, the equity gap was conceived as a supply-side market failure. Successive official reports in the UK (Bolton, 1971; Wilson, 1980; National Economic Development Office, 1986; Williams, 1998; Pickering, 2002; HM Treasury, 2003) over the three quarters of a century since the Macmillan Report have broadly echoed its findings that the capital markets are frequently discriminatory against smaller firms⁹. However, the phenomenon is not unique to one nation but universal to market-based economies obliged to make judgments on partial information.

It is perhaps not the longevity of the gap that is surprising but, rather, its continuing notoriety. That the gap remains an issue of substantive debate (HM Treasury, 2003) is in large part because of the changing nature of the enterprises affected by capital constraints. The financing problems experienced by firms which could be classified as “high tech” or “R&D intensive” (Butchart, 1987; OECD, 1997) only gained visibility in the latter quarter of the 20th Century. The use of the term ‘knowledge economy’ with its implication of intangible (and thus un-bankable) intellectual assets is similarly recent (Sweeney, 1977).

Murray (1995) and latterly Sohl (1999) have both argued that the use of the term “equity gap” in the singular is a misrepresentation of the harsher realities faced by the young and growing firm in its vulnerable years prior to the accumulation of sufficient collateral-based assets or reputation. They both suggest that there exists a second equity gap represented at the stage where seed or start-up capital had been exhausted and no additional providers were prepared to “follow-on” from the original external investor. For small early-stage venture capital funds or business angels with limited resources to fund follow-on rounds without the participation of new syndicate partners, the absence of external co-funding also severely prejudices investment performance.

This discussion implies the delineation of a range of funding which is considered to be within the equity gap problem. The UK government believes that the gap exists for small firms seeking investments broadly between £500 000 to £2 million (HM Treasury, 2003). Yet, its exact quantification remains vague and inconclusive. The term, and its estimation, is frequently anecdotal. More than one gap has been identified for more than one reason (Lawton, 2002; Sohl, 2003). The institutional venture capital industry largely denies the import of the gap arguing that there are few supply-side constraints. Rather, they counter that the (demand-side) failure is in the quality of the entrepreneurs seeking risk capital (Queen, 2002).

Causes of market failure when investing in knowledge-based industries

The term equity gap nicely describes a financing constraint affecting high potential but as yet immature and vulnerable businesses. The actual reasons causing investors to provide

insufficient finance are embedded in the investment process and the risks and reward that such investment decisions will incur. Further, there is an operational question of materiality. The investment process has high sunk costs and often little advantages of scale. Thus, small investments may incur transactions costs out of all kilter with the probable benefits of the investment. In these circumstances, a rational and informed decision not to invest in an early stage venture cannot be seen as a market failure. On the contrary, it is the market working effectively. None the less, there are genuine sources of market failure affecting new knowledge based firms. Two are particularly pernicious: information asymmetries and R&D spill-overs.

Information asymmetries

In extremis, a highly innovative but immature technology employed to produce novel products and services provided to new customers by a technically knowledgeable but commercially inexperienced entrepreneur (possibly coming from a university environment) and who is starting a new enterprise provides a full spectrum of the sources of potential risk that the venture capital investor has to manage (NEFI, 2005). At its earliest stages, the technology is not proven in its applications. Even if the technology works as it is envisaged, it will be used to create products and services which are not yet widely available nor in some cases even fully comprehended by either future suppliers or users. In such circumstances, how does the firm or its investor(s) determine the attractiveness of products or services that as yet do not exist? These information challenges will remain while the company grows (see Figure 5.1) up until that extremely unlikely event that the technology attains a dominant position and becomes comprehensively understood as an industry standard.

Insert Figure 5.1 about here

Thus, we can have simultaneously technology risk, market risk, managerial risk and financial risk each impacting on a new high-tech enterprise (Amit et al., 1990; Storey and Tether, 1998). Multiple decisions have to be made, often very quickly, on highly imperfect knowledge. As Amit et al. (1990) contend, less able entrepreneurs will choose to involve venture capitalists, whereas the more profitable ventures will be developed without external participation because of the adverse selection problem associated with asymmetric information. Amit's argument implausibly assumes that unknown entrepreneurs regardless of skills have alternative and sufficient sources of finance available.

Risk is a computable state based on estimated probabilities. Thus, seed, start-up and other early-stage investments in unique enterprises where no prior history exists are particularly problematic. Without reference benchmarks, investors also face incomputable uncertainty of a Knightian nature (Knight, 1921). Audretsch (2004) in referring to new technology environments uses the term "hyper uncertainty". The use of quantitative approaches is effectively nullified in such a speculative and volatile environment. The main implication of this situation is that the presence of non-quantifiable uncertainty affects commercial decisions by amplifying their perceived risk components (Einhorn and Hogarth, 1985; Kahn and Sarin, 1988; Ghosh and Ray, 1997). As a result, early-stage venture capital investments may offer investors the prospect of little confidence of higher returns but with a considerable likelihood of project failure. In such circumstances, the abandonment of seed investments in favor of later stage deals by commercial investors can be viewed as highly rational (Dimov and Murray, 2006).

R&D spill-overs

Audretsch (2004) also observes that it cannot be assumed that desirable spill-over effects, i.e. whereby society at large gains access to and benefits from the availability of a valuable new innovation, are automatic. The entrepreneur is a critical agent in the dissemination of innovative ideas. In early-stage classic venture capital activity, a majority of investments in a portfolio will either fail or return (at best) a negligible net present value when the time cost of money and an appropriate risk premia are computed

(Fenn et al., 1995; Murray and Marriott, 1998; Rosa and Raade, 2006). Where attractive net returns are made by the fund, it is likely to result from the realization of a small minority of exceptional investments within the portfolio (Huntsman and Hoban, 1980; Bürgel 2000). Given these uncertainties, the venture capital investors will seek to ensure contractually that when abnormal rents are generated they are owned by the investors (Van Osnabrugge, 1999). Hence, the attention given by technology investors to ensure that they have strong patent protection (Salhman, 1990; Kortum and Lerner, 2000). Indeed, the investors' ability to appropriate the commercial benefits of their actions is likely to affect their original investment decision. None the less, the full value of a novel technology and the consequent stream of new products and services are rarely harvested in their entirety by the investors. Competitors may emulate and copy key attributes, both legally and illegally. The bargaining power of suppliers or customers may also erode the innovator's surplus rents (Griliches, 1992). In a study of 600 high tech start-ups in Germany and the UK, Bürgel et al. (2004) found that the high-tech young firms experienced their first serious competitive threat after a median period of 16 months of sales.

In these circumstances, both entrepreneurs and investors may well feel that the enterprise risks and uncertainties are too high and their ability to secure both full and attractive returns from successful technology enterprises are too doubtful. This is likely to result in an undersupply of investment regardless of the fact that the existence of the innovation has benefits to a wide range of parties. Griliches estimates that the gap between the private and the social rate of return spans 50-100 per cent of the private rate of return. Small firms because of their lesser market power and inability to finance the aggressive defense of intellectual ownership infringements are particularly likely to see an erosion of their returns (Mansfield et al., 1977).

POLICY CHALLENGES IN THE VENTURE CAPITAL ARENA

Based on a near universal admiration as to the vigor of the US innovation financing system, several governments have sought to emulate elements of the American venture capital system. The assumption is made, often implicitly, that elements of a system may

be isolated and applied within other different contexts. This raises a set of issues of both theoretical and operational complexity that policy makers ignore at their peril.

The influence of a US exemplar

Given the noted hegemony of the USA innovation finance system, it is legitimate to question what can be learned from the American experience of venture capital activity and be readily applied to other national economies both in the developed and developing world. Yet the simplicity of the question betrays an ignorance of both “path dependency” (Kenney and von Burg, 1999) or what it is that can actually be made transferable even assuming that the environmental and institutional conditions (e.g. tax regimes, legal and corporate governance structures, etc.) exist for such a transfer to be possible¹⁰. Gilson (2003) is explicit in his assertion that others cannot follow USA experience in order to reach “the holy grail” of a flourishing venture capital market modeled on Silicon Valley. He notes that because of the US industry’s highly idiosyncratic history, “the manner in which the US venture capital market developed is not duplicable elsewhere” (p. 3). He goes on to argue that other countries might be obliged to use public policy measures given the inability to copy another country’s history¹¹.

State interest in entrepreneurial action has moved from exhortation to involvement as many commercial investors have abandoned early-stage equity finance (Sohl, 2003; Cummings et al., 2005; Collier Capital, 2006). General and limited partners’ actions are an articulate judgment of the economic attractiveness of the early-stage market. Their desertion has left a financing (equity) gap that governments have felt obliged to try and fill. This move from early to later stage deals (“style drift”) is most evident in European venture capital markets. However, it is not an exclusively European phenomenon. Gompers (1998) shows that US investors also moved to later stage deals as the size of the finance under management by venture capital general partnerships increased rapidly in the late 1990s. This same phenomenon was earlier described by Bygrave and Timmons (1992) and more recently by Branscomb and Auerswald (2003).

The specific problem of minimum fund scale

“Ask an LP what he thinks of investing in European venture tech and he is likely to respond “what is European venture tech?”

(European Venture Capital Journal, November 2004)

The single biggest problem – facing both governments keen to encourage early-stage investment in new technology based firms as well as for general and limited partnerships prepared to consider early-stage risk capital investment activity – is simply put. With very few exceptions, the investment record of early-stage funds world-wide has been very poor (European Venture Capital Association, 2005). The general exceptions to this rule over the long term have been from the upper quartile of US technology investors. The consistency of poor venture capital returns in Europe has been so uniform as to make a number of institutions question whether Europe actually has a viable, early-stage technology investment activity (Ernst and Young, 2004).

Insert Table 5.1 about here

The pooled IRR statistic of All Ventures for Europe of 6.3 per cent p.a. in Table 5.1 can be compared to the US equivalent of 16.5 per cent p.a. for the period 1986-2006. (Both sets of statistics are from Thomson.). Söderblom and Wiklund (2005) in seeking to determine robust reasons for the apparent consistent difference in performance between US and European early-stage venture capital funds reviewed over 120 academic papers. They concluded that the following venture capital firm related variables appeared to be repeatedly associated with successful professional equity investments:

- Industry specialisation (knowledge effects);
- large fund size (scale effects);
- strong syndicated deal flow (network effects);
- management and technical competence (human capital effects); and

- large and rapid investments per successful enterprise (implementation effects)

Collectively, these firm level influences can be interpreted as the positive application of “scale and scope economies” to the risk capital investment process.

Insert Figure 5.2 about here

Murray and Marriott (1998) in a simulation of early-stage venture capital fund activity showed that fixed costs have a severe effect on the net performance of small funds. Excess costs particularly fell on the venture capital firm or general partner (Figure 5.2). This view is also corroborated by Dimov and Murray’s (2006) analysis of the supply determinants of seed capital in their investigation of seed capital fund activities from 1962 to 2002. They showed that the most active seed investors over time were almost exclusively large and well established US funds. The top five venture capital investors active in seed capital which were all US based¹² had, on average, made 92 seed investments from total funds under management of nearly \$4 billion per venture capital firm.

Small is *not* beautiful

The most conspicuous outcome of these studies is to challenge the apparent willingness of policy makers to create and to support programs which encourage the formation of sub-optimal, and ultimately non-viable, seed capital funds. Here, the state can be seen as both part of the problem as well as a possible solution to financing constraints. In the state’s absence, few private investors would have participated in such funds. Small, early-stage fund have a series of structural weaknesses that serve to undermine the probabilities that these funds will earn an acceptable risk adjusted return on the finance under their management (Figure 5.3). One effect of this sub-optimal fund size is a particularly

damaging inability to recruit experienced professional investment executives in a highly competitive market for talent (European Investment Fund, 2005).

Insert Figure 5.3 about here

Insufficient fund size similarly reduces its attraction to institutional investors. These investors including pension funds, insurance companies and other money managers become limited partners in venture capital funds in order to add some controlled exposure to high risk/high reward opportunities (Bürgel, 2000; BVCA 2000). Because of the multi-billion dollar global reach of these institutional investors, asset allocation has to be material in order to have any effect on the performance of their investment portfolio. In these circumstances, a minority position for a limited partner demands involvement in a fund of sufficient scale in order to influence the institution's overall performance. For all but the smallest institutional investors, a minority position in a closed venture capital fund of under US\$100 million is likely to be irrelevant.

Thus, a small seed fund's circumstances often describe the worst of all worlds. It has modest funds to invest and little income with which to execute a strategy of finding and evaluating potential investee firms. It is unlikely to be fully diversified. Investee firms, particularly at start-up, are costly in their urgent need for advice and the fund has little money to provide follow-on finance for the most attractive prospects in its portfolio. The lack of finance results in either impeding the portfolio firm's growth plans and/or in the rapid dilution of the fund's ownership share as syndication finance is arranged. These conditions are very likely to lead to a sustained poor investment record which will severely reduce the fund's ability to survive by attracting subsequent follow-on funds from private sector investors. In these circumstances, policy makers outside the elite technology clusters of the USA, are likely to see the earliest and most uncertain stages of equity investment virtually abandoned by commercial venture capital firms. Governments are obliged to come to a view as to their own response to such a withdrawal of private, early-stage funding sources to priority (new technology) young enterprises. Almost

universally, they have considered the reduction (or absence) of support for such firms to be strategically and politically unacceptable.

GOVERNMENT INSTRUMENTS TO PROMOTE INSTITUTIONAL VENTURE CAPITAL FINANCE

Government's influences on the entrepreneurial vigor of a national economy are manifold. For example, the institutional legal and fiscal frameworks (Fenn et al., 1995; La Porta et al., 1997; 1998), the incentive structure of personal and corporate tax systems, the regulatory regime's impact on business, and the efficiency of the market for corporate control will each have profound (albeit not easily predicted) effects on the incentives for entrepreneurial risk taking. As such, these influences also affect the demand for venture capital as an important source of entrepreneurial finance.

Insert Figure 5.4 about here

The eclectic nature of popular policy recommendations reflects the wide spectrum of important influences on venture capital activity. Further, such conditions in order to be effective will in turn have to be embedded in, and legitimized by, a culture of opportunity recognition and entrepreneurial endeavor (Shane, 2003). However, given the dynamic and linked nature of investment activities to both micro- and macro-economic variables, it is not necessarily easy to determine what factors are most important at any one time. Factors that constrain entrepreneurial activity (e.g. legal and regulatory compliance) may not by their removal necessarily act as an accelerator for greater entrepreneurial endeavor.

A full treatment of environmental conditions conducive to venture capital finance is inappropriate in this chapter. Accordingly, we will focus specifically on what measures the state may employ *directly* to support the actions and effectiveness of both the institutional and informal venture capital industries.

The growing status of entrepreneurial activity

The late 20th Century saw a coming together of two related trends of 1) an increasing awareness of the importance of small and young firms to the wider economy after Birch's seminal MIT study (1979), and 2) an appreciation of the changing nature of economic value as represented by the growing importance of innovation via knowledge-based goods and services in mature and developed economies (Nonaka, 1994; Grant, 1996; Spender, 1996). Young high-tech firms were increasingly seen as an important conduit for continuing innovation particularly in fostering disruptive and non-incremental, technology developments (Druker, 1985; Storey and Tether, 1998; Audretsch and Zacs, 1990; Audretsch, 2002; Branscomb and Auerswald, 2003). Incumbent large firms were all too often seen as reactionary and thus vulnerable to adept new competitors (Christensen, 1997). The confluence of these new understandings meant that it was highly unlikely that governments could envisage not supporting entrepreneurial young firms. The "political stock" of small firms increased throughout the 1980s and 1990s. The technology and internet-related bull markets of the latter part of the 1990s, with their focus on young knowledge-based firms, further fuelled public and government interest in entrepreneurial activity. Accordingly, the specific financial constraints faced by young firms in their attempts to grow, and particularly the appropriateness of risk capital finance to high potential, new enterprises, became an increasing focus of policy interest.

Despite setbacks, there is also evidence that governments can and do learn over time. The environmental preconditions to effective entrepreneurial action including its financing are increasingly being recognized in policy circles (OECD 2004; Small Business Service, 2005; US Department of Commerce and European Commission Directorate General for Enterprise and Industry, 2005). Thurik (2003) summarizes Europe's eclectic policy needs in order to stimulate greater entrepreneurial activity:

Insert Figure 5.5 about here

Thurik's span of policy prescriptions reflects closely both the EC's 2003 document *Green Paper: Entrepreneurship in Europe* and similarly the UK government's own 2004 policy roadmap for a more entrepreneurial Britain (Figure 5.6). These enabling environmental conditions set a context in which venture capital programs can operate. The UK model is interesting for its belief in entrepreneurial activity and its comprehensive linking to government departments with both a commercial (e.g. DTI, Treasury and Inland Revenue) as well as a social mandate (Home Office). The encouragement of new enterprises in economically disadvantaged communities borrows from earlier experiences of immigration into both the British and US economies.

Insert Figure 5.6 about here

The two above illustrations raise an important issue of policy priorities and focus. Venture capital is an important instrument for promoting enterprise¹³. But as the US experience has also shown us, risk capital has played a critical role in the emergence of technology hubs on the East and West coasts. Thus, venture capital is additionally seen as a major instrument of Innovation policy. The conflation of the two policy goals is likely to result in some contradictions. Entrepreneurship is largely about mass activity including a wider interest in new enterprise at all levels of the citizenry. In contrast, innovation policy is frequently much more meritocratic in nature and seeks the promotion of technologies and enterprises of world-class competitive status. However, as the "TrendChart Innovation Policy in Europe" (European Commission, 2004) report notes, priorities on innovation among EU states include "fostering an innovation culture" and "building innovative capacity in smaller enterprises." These instruments and goals of both innovation and entrepreneurship policies often appear remarkably similar.

Venture capital intervention typologies

As noted earlier in this chapter, governments have restricted their direct financial involvement in venture capital to the more problematic investment stages of seed, start-up and early firm growth (OECD, 2004). Their interventions are premised on a belief of the key role that professional risk capital can make to the innovative capacity of an economy. Hence their near exclusive focus on supporting seed, start-up and early growth stages. In these earliest and most speculative stages, it is still not fully evident that specialist early-stage venture capital investment is an activity than can be effectively mediated through a market mechanism as apposed to directed grants or subsidy alternatives. As noted, this uncertainty is exacerbated by the historically poor returns to early-stage venture capital activities. Outside the special case of the United States “high tech” industries prior to the year 2000, risk-return characteristics have consistently been unfavorable for investors wishing to engage in early-stage ventures (Murray and Lott, 1995; Murray and Marriott, 1998; Lockett et al., 2002; Rosa and Raade, 2006)¹⁴. In direct contrast, the later stages of venture capital and private equity have been consistently profitable with management buy-outs becoming the European industries’ dominant product (Wright et al., 1994; EVCA, 2005). Private equity does not exhibit comparable problems of insufficient scale or information asymmetries. Thus, government involvement in such later-stage actions is rare other than in the setting of the appropriate enabling legislation (Wright and Robbie, 1998).

Having made the decision to intervene in the market for early-stage venture capital, the state has to decide what form of intervention for what purpose will be most effective. While there are a number of national studies of venture capital programs (Lerner, 1999; Dossani and Kenney, 2002; Maula and Murray, 2003; Ayayi, 2004; Lerner et al., 2005), the value of such precedents is in part obscured by the specificity of the programs to their national context (Jääskeläinen et al., 2005). None the less, the risk capital decisions of government can be pared down to two generic choices:

1) *Direct Intervention* – government as venture capitalist

Government may elect to become its own venture capitalist and undertakes all the roles that would otherwise be the responsibility of a rent-seeking, market intermediary. Here, the government run, venture capital firm has to undertake all the selection and due

diligence, governance and nurturing duties incumbent on an early-stage risk capital investor. Given that the government is investing public finance in order to fulfill its role, the state assumes simultaneously both the roles of general and limited partner in the public fund.

2) *Indirect Intervention* – private venture capital firms as agent of government

Alternatively, government can delegate executive responsibility to a private venture capitalist fund manager. This is often done on the assumption that the state has neither the professional skills nor the experience to be a “direct” risk capital investor. Once operational responsibility has been assumed by a private general partner agent, the position of the state becomes analogous to that of a limited partner in a traditional limited liability partnership (LLP) fund. Limited partners are not able to intervene in the operations of the fund manager at the risk of losing tax status privileges. Similarly, the government’s operational involvement ceases once the focus and mode of operation of the fund has been agreed and public monies committed. Indirect intervention via equity enhancement schemes is becoming the dominant contemporary mode of public involvement as the importance of highly skilled, and properly incentivised, investment managers is recognized (Gilson, 2003; OECD, 2004).

Government uses multiple incentives with which to encourage the involvement of private venture capital agents. It will exploit the power of the state to provide additional and cheaper finance to the fund thereby allowing a leverage effect to increase “up-side returns”. Government may further reduce the costs of the venture capital fund by contributing to a proportion of the operating costs of the fund. However, such operating subsidies are less common than alternative measures with a direct incentivising effect on the fund’s performance. Finally, the state may change the risk/reward profile of the fund by underwriting part or all of the risk of financial loss to the limited and general partners.

Direct public involvement

The state as a direct investor creates some challenging issues. First, there is the question as to whether government has appropriate personnel capable of carry out such commercially sophisticated activities as equity investment in early-stage firms. It is unlikely that such persons are widely available as civil servants. Secondly, the state by its size and influence inevitably will create an impact – for good or ill. Thirdly, that the state has to assume the role of a direct investor may be seen as a consequence of its failure to incentivise a private market to take on this largely commercial role. A number of countries do have direct investment via public bodies. The Finnish Investment Industry program with civil servants investing public money directly into young enterprises primarily to fulfill government policy goals would meet this definition (Maula and Murray, 2003). Similarly, the Danish government financed *VækstFonden* (Business Development Finance) is also has a direct venture capital investment activity¹⁵. Yet, these programs raise some very considerable issues regarding the conflict between policy and commercial goals.

As noted, direct involvement in new venture investment by government agencies carries a material risk of market disruption through the potential misallocation of capital and the possible “crowding out” of private investors (Leleux and Surlemont, 2003). These undesired effects can be due to both the commercial involvement of inexperienced public service personnel, who may often control substantial levels of finance relative to the total supply of early-stage venture finance in a market. In addition, there may be differing return requirements of public investors (OECD, 1997; Manigart, et al., 2002; Armour and Cumming, 2006). Government funding can further distort private markets by offering finance which does not fully reflect the appropriate risk premium (Maula and Murray, 2003). Venture capital is a “learned” activity (Bergemann and Hege, 1997; Shepherd et al., 2000 and 2007; De Clercq and Sapienza, 2005). General partnerships often need the experience of managing and investing multiple funds before they have accumulated sufficient technical and market knowledge to provide their investors with acceptable returns (Gompers and Lerner, 1998). Many public programs have recognized the possible adverse effects of government inexperience on market outcomes. None the less, government’s direct intervention in the supply of venture capital has frequently been defended on market failure arguments without reference to the efficacy of such actions. It

is perhaps inevitable that criticisms of market distortion have been leveled at such public programs. For example, the Canadian Labor-Sponsored Venture Capital Corporations – a program with both commercial and social goals and made attractive to individual investors via substantial federal tax breaks – have been accused of crowding out private venture capital activities (Cumming and MacIntosh, 2002).

Indirect or ‘hybrid’ public/private venture capital models

The OECD has used the term “hybrid” to describe the structures where government and private interests work in concert as co-investors, i.e. limited partners, in a fund. Such structures are seen as an element of “best practice” (OECD, 1997; 2004; European Commission and US Government, 2005). In effect, the venture capital firm or general partner is acting as an “agent”¹⁶ for a group of principals (limited partners) one of which is the state using public monies. Governments’ history as active investors selecting commercially attractive projects is problematic at the very least with many illustrations of adverse selection. “Spotting winners” among young and growing companies is fraught with danger (Hackim, 1989) and denies the stochastic nature of the firm formation process. A general conclusion from the last half a century is that government would do well to avoid placing itself in a position where it has to make nominally economic decisions that are bounded by other, usually political, conditions (OECD, 1997; Modena, 2002; Gilson, 2003). There appears to be a growing consensus on the limited role of government as a direct investor. Contemporary venture capital programs in, for example, the USA, the UK, Australia, New Zealand and Germany have each used private venture capital firms to invest on behalf of government in areas of new enterprise deemed important for policy reasons. The parallel involvement of both public and private investors can be seen in Figure 5.7.

Insert Figure 5.7 about here

That the state would wish the participation of private investors to support its policy goals is self evident. But why commercial investors would be prepared to be involved as limited partners in such a hybrid activity needs further explanation. Such an arrangement may do little to alter expected outcomes that led to the supply side, market failure in the first place. Thus, the involvement of the GP and any private sector LPs in the fund will frequently require the engineering of more attractive profit expectations in order for them to participate (Gilson, 2003; Maula and Murray, 2003; Hirsch, 2005; Jääskeläinen et al., 2005).

In discussing the evolution of different incentive structures in government's support of venture capital, it is important to acknowledge the contribution of the Small Business Investment Company program created by the US Government's Small Business Administration¹⁷. The basic model of an "equity enhancement" program by which the state's involvement (either by direct investment or acting as a guarantor to other fund providers) enables additional and cheaper funds to be raised – thereby creating a leverage advantage to private investors – has been reflected in venture capital programs world-wide.

It should be stressed that the full value of the Small Business Investment Company program in its various forms (i.e. bank owned, debenture and participating securities funds) was not restricted to the net returns generated by the funds. Indeed, the investment performance of these funds has been highly variable over time (Small Business Administration, 2002; 2004). Rather, the value of the program has been in the espousing of novel public-private experiments used to address the problem of the inadequate supplies of risk capital for young firms across a range of communities. Of critical importance, the Small Business Investment Company program also enabled a generation of US investment managers to obtain their first commercial experience of venture capital activity via government supported funds. There is an "infant industry" argument for interceding in immature markets (Baldwin, 1969; Irwin and Klenow, 1994). In the UK, the government conceived venture capital firm, 3i plc and its predecessor ICFC, performed a similar industry development role from 1946 until the late 1980s (Coopey and Clarke, 1995).

Public funded incentives in hybrid funds

Hybrid funds illustrate the imperative of government action in strategically important investment categories where consistently poor returns have precipitated a major reduction in private finance for young enterprises. Such structures also acknowledge that good investment managers will rarely tolerate the state having an executive role in their funds. This impasse is resolved by the use of government finance to incentivise private managers to engage in more risky, early-stage funds. It is the “hands-off” provision of government leverage finance that has appealed to professional investment managers mindful of the need to increase fund scale in the challenging early-stage markets and frequently faced with private investor indifference.

The term “equity enhancement” is accurate. The state needs to incentivise the general and limited partners to be prepared to engage in a fund that includes public welfare goals as part of its *raison d’être*. In the absence of explicit economic incentives, there is little logic for profit maximizing, private agents to become involved. *Pari passu* funding, whereby the state and private investors provide investment finance on equal terms, only works where the returns to private investors are attractive enough without needing to skew the return distributions to private limited partners’ advantage. The state is obliged to enhance the returns to the general partner and to the other commercially motivated limited partners in order to attract private investors’ commitment to the co-investment model. Essentially, the public investor forgoes some of its rights to a *pro rata* share in the economic returns of the fund. Given that the state is often the largest single investor in the fund, a reduction of its share of any net capital gain can leverage a material increase to the other parties’ returns in the event of a moderately successful fund. In all cases, the state as a “special” limited partner does not influence the commercial and executive decisions of the fund managers once the investment criteria of the fund are determined. Relative distributions of surplus are agreed *ex ante*. Hybrid funds commonly employ one or more of range of incentive structures:

- 1) Capped returns to the state

The public limited partner invests at a rate which is commonly fixed at approximately the government's cost of capital¹⁸. Once this return threshold has been met from the proceeds of the sale of any portfolio companies, any further positive cash flows are exclusively shared between the other commercial LPs and the GP via its "carried interest".

2) Differential timing of limited partners' cash flows ("first in and last out")

For the general partners of the venture capital firm, their performance will be largely measured and communicated by one universal metric – the Internal Rate of Return of the fund¹⁹. When the state is the first investor to have its committed finance fully drawn down and the last investor to have its monies returned with any associated surplus, the shorter investment period of the other private LPs directly benefits their returns. Again, the performance enhancement of the private or commercial partners is at the direct cost of the public partner.

3) Guaranteed underwriting of investment losses incurred by the limited partners

Government may seek to influence the investment decision by removing all or a large part of the costs of portfolio failure. Usually, a percentage of committed investments is guaranteed which may often vary from 50-75 per cent of investors' costs. The guarantee may be on a fund or on individual portfolio investments. However, the guarantee schemes do not, in themselves, improve the returns to investors but rather cap losses. Thus, it is common for a guarantee to be put in place in addition to some other incentive which leverages the upside of a successful investment.

4) Buy-back options

Buy-back options give the private investors the opportunity to purchase the entirety of government's interest in a program at a pre-determined time during the public/private program. The terms of the exchange are arranged *ex ante* and thus present the private investors with a clear incentive and opportunity for an early exit of the state as co-investor. Essentially, the facility is a purchase "option" which may be exercised at some stipulated future date when economic future of the investment is possibly indicated but

not fully known. One of the most admired of such programs was the Israeli Yozma program (1993-98). Seven countries have subsequently modeled programs on this Israeli experience.²⁰

A list a number of existing government supported venture capital structures that have adopted one or more of the four described incentives programs is given in Table 5.2.

Insert Table 5.2 about here

Despite the increasing popularity of government involvement in hybrid, venture capital funds, rigorous evaluation of their effects is limited. Evaluations have been undertaken on such schemes in the UK, New Zealand and Australia²¹. However, unabridged evaluations are rarely available in the public domain²². Thus, while we may assume that subsequent program rounds have learned from policy experience, our overall knowledge of contemporary venture capital policy actions and outcomes is limited. Agreed evaluation methodologies and the ability to compare and contrast across program and country are each urgently required in an area of government interest and action of increasing scale (Lundström and Stevenson, 2005).

Government's role in venture capital 'funds of funds'

The hybrid venture capital fund structure assumes that government is the co-partner and limited partner to an individual fund. In the "fund of funds" option, the government does not signal preference for any one fund but supports investments in a range of funds that are sanctioned by the general partner management. The fund of funds manager, acting as an allocator of limited partners' commitments, allocates finance to specific venture capital funds not to individual investments. While several such fund of funds exist, governments tend to invest only in such groupings that specifically target young and high potential firms of policy interest. In France, the Fund for the Promotion of Venture

capital (*Fonds de Promotion pour le Capital Risque* – FPCR) was set up in 2001. With Euro150 million, including European Investment Bank support, at its disposal, the FPCR has invested in 10 French venture capital funds. Investments are geared towards innovating companies less than 7 years old in sectors where it is difficult to mobilize private funding, i.e. life sciences, ICT, electronics, new materials and environment and sustainable development. The UK equivalent is the UK High Technology Fund. This fund of funds, set up by government in 2000 and managed by a commercial investment company, has raised £126 million of which £20 million only came from government. It invests exclusively in specialist technology venture capital funds located in the UK. This fund of funds also attracted EC assistance via the co-investment of the European Investment Fund. It is yet too early to appraise the performance of either of these programs.

GOVERNMENT POLICY AND INFORMAL INVESTORS (BUSINESS ANGELS)

So far in this chapter, discussions have focused on policy actions in the institutional market for risk capital. Thus, the unit of analysis has been the established, and often very visible, venture capital firms or general partnerships. Yet, advocates of the importance of business angels have repeatedly noted that the scale of the informal supply is likely to dwarf that of the institutional venture capital in all developed risk capital markets. There is clear evidence of this disparity in the US and the UK where both types of investor co-exist. This argument regarding the scale, and thus potential for economic transformation via business angels, is strongly made and mutually re-enforced by authors Mason, Kelly, Riding, Madill, Haines and Sohl writing in this book. Accordingly, governments concerned at the effective supply of financial and business related support to young and growing companies have increasingly become interested in informal investors or business angels²³. A Kelly rightly notes, government interest in these “publicly hidden” investors was materially influenced by the work of pioneering academic researchers both in the USA and Europe.

Given that three chapters in this book are devoted to different aspects of business angel research and practice, this author will not repeat the analysis of acknowledged experts in the field. Rather this section will remain exclusively within the policy focus of the chapter and will look at how governments have sought to promote a vigorous and successful business angel sector.

Investors of first resort

Van Osnabrugge (1999) “guesstimates” that business angels provide between 2-5 times the amount of finance to entrepreneurial firms in the US and possibly invest in up to 40 times the number of portfolio companies compared to institutional venture capital firms. Bygrave et al. (2003), using GEM data on 15 nations, subsequently offer more modest differences in business angels’ favor of 1:1.6 re funds allocated. Sohl (1999) argues that the ratio of understanding of business angels compared to their *impact* on the economy is lower than just about any other major contributor. Bygrave et al. (op. cit.) support this view noting that entrepreneurs – and policy makers – should pay far less attention to institutional venture capital activities. They observe that new enterprises, including those involved in the commercialization of revolutionary research, are much more likely to be self-financed or gain support from business angels rather than from classic venture capital firms. In an international venture capital and private equity industry which is becoming increasingly dominated by scale, those parochial investors that are prepared to undertake local investments within the equity gap spectrum are an increasingly valued asset. The overall trend is for an increase in size, and thus concentration of venture capital funds, that continues to militate strongly against small investments²⁴.

A complement to institutional investors

In an ideal policy maker’s world, one might envisage informed and highly experienced business angels being the predominant seed capital providers to nascent businesses. Through their own commercial experience in allied sectors or activity, they would be able to offer the new enterprise valuable practical experience of running a young company. Of

equal value, they can provide a “certification effect” allowing fledgling entrepreneurs access to valuable commercial networks (Birley, 1985; Stuart et al., 1999; Steier and Greenwood, 2000), as well as offering timely and pertinent advice on markets, production etc. As the investee firm starts to grow and to demonstrate a major economic opportunity, the business angel could be the conduit to institutional venture capital. With firm growth accelerating, the influence of the business angel would give way to the more structured and “enterprise nurturing” skills of the venture capital firm (Harrison and Mason, 2000).

However, as with formal investing, practice is likely to fall short of idealized expectations. The business angel invests his/her own money and thus does not need to meet external regulatory standards. The venture capital firms are handling investors’ monies and must therefore be registered with the appropriate financial regulators. The former can act as idiosyncratically as they wish. Hunch, gut feel and subjective stimuli are all found to be important investment decision criteria. The institutional venture capital manager is much more likely to undertake an industry specific training and to have clear guidelines from both the general partners of the fund as well as industry guidance from the national venture capital industry association on due diligence, deal pricing, use of share structures, legal contracts etc. Accordingly, unless well known and trusted by the venture capital investor, the idiosyncratic and untrained business angel is likely to be viewed with mistrust as an investment syndicate partner. As Kelly wisely notes: “complementarity” does not presume “compatibility”.

Business angels as a policy focus

The three writers on business angels in this volume each make some brief observation on the policy implications of business angel activity. Mason looking at the spatial dimensions of investment argues that a supply of attractive investments will generate the supply-side response of adequate investments funds. His argument strongly reflects the venture capital communities’ argument that effective demand (i.e. quality of proposals) is the greatest constraint to early-stage financing. Perhaps more interestingly Mason and Harrison (2003) have argued that the UK government, by promoting the Regional

Venture Capital Funds program to address both spatial and early-stage inequities, have misunderstood both the nature of the problem and its prescription. They make a telling argument that business angel finance could be a much more appropriate response to such problems. Kelly mentions the three related problems of an excess demand for business angels' equity finance from would-be entrepreneurs; the information asymmetry or search problem of investors and firms not knowing that each other exists; and the incentives problem of getting 'virgin' business angels to turn intention into tangible investment activity. Finally, Sohl, addresses four areas of policy including promoting better linkages, sponsoring research in the field, more education of business angels, and unambiguous incentives for business angels to take the risks of equity finance. How government has actually sought to tackle the issue of stimulating informal investment will be addressed in the following sections.

Targeting business angels as individuals

Government has generally adopted a two-pronged strategy to facilitate the supply of informal investors in the national and local market. Firstly, they have had to address the "indirect measures" (OECD, 2004) of taxation in order to ensure that the incentives available to private investors are commensurate with the level of (undiversified) risk that they have to assume. The major suppliers of venture capital at the institutional level are frequently tax exempt in the most developed venture capital economies, i.e. pension funds, insurance companies and university or family trusts²⁵. But for informal investors, the means by which successful investments are made and recouped are profoundly influenced by the detail of the prevailing personal taxation regimes. The OECD's view is that it is counter-productive for a private investor to incur such high taxation and other costs that they reduce the underlying logic for making the original investment. It is difficult to argue with this orthodoxy.

A number of countries have "front end" incentives that give significant income tax shelters for bearing the cost of entrepreneurial activity. In the event of a successful investment realization, Capital Gains Tax may be delayed or removed from investments held for specific lengths of time. Such fiscal incentives exist in, for example, the US, the

UK, France, Ireland, Belgium and Canada. Because business angels are usually relying on their own personal income and wealth for investment activity, the system seeks to incentivise them to remain active investors by improving (and protecting) their returns compared to those parties not involved in the incentive schemes. However, as the OECD also acknowledges, while sophisticated changes to the tax system for high net worth individuals may prompt them to make more investments and/or retain their investments in growing companies for a longer period, it may also confuse other less sophisticated, informal investors.

The UK has been one of the developed economies most interested in using fiscal incentives to encourage “virgin” angels. In 1981, the Business Start-Up Scheme was announced. This program, which was the first to offer entrepreneurial investor incentives, evolved over time to become the Business Enterprise Scheme. This was established to enable investors to obtain tax relief when purchasing ordinary shares in unquoted firms seeking seed-corn funds for development²⁶. It ran ten years from 1983 to 1993. In turn, the Business Enterprise Scheme metamorphosed into the Enterprise Investment Scheme in 1994²⁷ in order to provide a revised program that would incentivise individual investors to provide more risk capital funds for the UK’s SME sector²⁸. In a review of the efficacy of this latter and extant program (Boyns et al., 2003), its authors concluded that the schemes had created significant “additionality” thereby improving the resources available to young firms while increasing investors’ returns. The simplicity of the Enterprise Investment Scheme is particularly attractive to business angels as it makes no attempt to determine investment eligibility other than stipulating the qualifying and non-qualifying categories of investment.

Insert Figure 5.8 about here

The UK scheme is not unique but rather is more established than many similar programs. It may be seen as an archetype or model given that it replicates the key criteria evident in most programs, namely:

- Higher risk, young companies clearly targeted and specified by age, economic size and type of activity;
- total tax forgone is capped for both the recipient company and the individual investor;
- *ex ante* tax relief given on investors' income when equity investment made in target business;
- investment losses can be used in further personal tax computations; and
- *ex post* Capital Gains Taxes either avoided or reduced after a minimum holding period.

Based on several countries' experiences, a European-wide program, the Young Innovative Company Scheme is currently evolving (EuropaBio, 2006). It is proposed that, for eligible investments, no tax is levied on capital gains realized on stocks that have been held for a minimum of three years. The French equivalent of the Young Innovative Company program (i.e. *Jeune Entreprise Innovante*) was adopted by the French Parliament in the 2004 Budget Bill. A similar bill, HR 5198, the Access to Capital for Entrepreneurs Act of 2006, was presented to the US Congress in April of this year.

Targeting business angels as co-investors

These above schemes are directed at incentivising the individual to invest directly or via a tax efficient, trust fund structure. Government has also increasingly seen the business angel community as an entity that may be incentivised collectively at the fund level in a potentially similar fashion to institutional venture capital businesses. To date, the tax transparency attractions of the Limited Liability Partnership structure have largely been irrelevant to the private investor. Yet, a syndicate of business angels investing co-operatively on projects where the due diligence and other investment costs has been shared, as well as any eventual capital gains, is broadly equivalent to the established practices of the established venture capital industry. In recognition of this reality, the UK government has tried to find means by which it can invest alongside informal investors. Such leverage initiatives have had to address and accommodate the legal issue that the

business angel is a personal investor rather than an institutional investor. However, as business angel networks or bespoke syndicates start to manage external funds, these differences begin to blur.

A number of recent developments are noteworthy. Governments may invest as a public limited partner in a fund specifically made up of business angels investors. The UK's 2004 scheme, the Enterprise Capital Fund, is designed in such a manner as to accommodate both institutional and informal investor groups. The bringing of business angel investors into a legal identity as a group or fund is relatively novel. It addresses the common problem of business angel investors being severely curtailed in the size of personal investments. Historically more common, the state may be a co-investor at the level of individual projects. Such a scheme has been in operation for several years in a primarily institutional venture capital context in Germany via the BTU program. An informal investor equivalent is seen in New Zealand's Seed Co-investment Fund. This latter program that has been designed in the light of UK experience via the Scottish Co-Investment Fund and the London Business Angel program.

These schemes each bear common criteria of informal investor/government interaction:

- State acts as a co-investor increasing scale of investment available;
- once conditions of investment eligibility met, the state is passive;
- criteria of eligible investment defined by age, size and economic activity; and
- state's position as an investor is usually subordinate to private investors.

Networks, portals, match-makers and information asymmetries

The institutional venture capital industry, like many professional services, may be seen as a dense network of complementary actors associated in the common delivery of specialist financial products and sharing critical information. Over-lapping venture capital networks may, for example, be classified by stage of investment, types of opportunity, or location. Issues of access and preference determine a pecking order of venture capital firms and partners efficiently calibrated by industry reputations. The intensity of the venture capital

locations or clusters in a relatively small number of major cities across Europe, America or Asia further increases the ability to communicate effectively and quickly between interested parties. To date, the venture capital industries have been broadly characterized as country specific. However, as the venture capital industry matures and leading players grow and expand their locus of operations at the expense of less successful partnerships, the incidence of internationalisation has increased markedly (Mäkelä and Maula, 2005; Deloitte and Touche, 2006).

Communications between informal investors or business angels compares poorly to the small number of well organized venture capital firms located within the umbrella of an efficient and highly representative industry association. Gilson's (2002) "simultaneity problem" exists but is much more acute in an informal venture capital environment. A business angel has to signal to would-be investee businesses that he or she is interested in receiving business proposals. At the same time, the informal investor may wish to inform other angels of his/her²⁹ presence and willingness to participate in syndicated investments. This lack of preceding contact, the diverse personal histories of the investors and the absence of physical market places each serve to confound efficient communication. In the absence of such communication, it is similarly difficult to ensure that investment proposals are fairly appraised and sensibly priced. It is not uncommon for an inexperienced investor to face an equally inexperienced entrepreneur with neither party able to assess the quality or the credibility of the business proposal or the financing offered. Such circumstances are very vulnerable to inefficient (or disastrous) outcomes either by chance or design. It is for these reasons that many in the insitutional venture capital industry remain highly ambivalent as to the involvement of business angels in professional venture capital investment deals. The cost of sorting out badly constructed or poorly priced financial arrangements previously arranged between an inexperienced business angel and entrepreneur may be sufficiently time consuming for the venture capitalist to walk away from providing follow-on finance. As Gifford (1997) has noted, the constraining resource for many general partners is management time rather than the flow of opportunities or finance.

In these circumstances, the involvement of government in activities which increase the information available to investors and/or investees is likely to be productive at modest

cost. Similarly, the powerful national venture capital associations also have an interest in ensuring that informal investors looking for early-stage deals that might lead on to institutional venture capital activity are also sensibly advised, financed and structured. The policy benefits of reducing information asymmetries and ensuring deals are properly communicated to an active and deep market appear universally accepted. Accordingly, business angels networks on a local, national and (on occasions) international basis have been supported by both government and institutional venture capital associations. Sohl uses the generic term “portal” to describe this interface between investors and entrepreneurs. Public transfers have also allowed the networks to be sufficiently well financed to ensure the recruitment of appropriate management and training resources.

A dissenting voice

Lerner (1998) turns a refreshingly skeptical eye on business angel activity. This is a useful antidote as the business angel literature is often highly evangelical in its arguments. Lerner asks two questions that he contends are too often assumed rather than posed: 1) do private capital markets provide insufficient capital to new firms; and 2) can governments better identify future winning businesses in which to invest? In the absence of robust empirical examination³⁰ of both the contribution of angels and the value-added role of government, Lerner remains a skeptic. Picking up a major theme of this chapter, he notes that the importance of scale is clear but business angels, with very few exceptions, are commonly characterized by their modest investment resources. Even when banded together in investment syndicates, they are seldom likely to create an aggregate fund size of \$>50 million. In classic venture capital terms, such a small fund size would now widely be seen as uneconomic. Lerner’s concern is that new enterprises of insufficient potential to be financed by institutional venture capital firms are then taken up by sub-optimally sized business angel networks where the range and level of investment skills and experience are highly variable. In effect, the informal investor network is both “second best” in its resources and in the potential assistance that it can offer entrepreneurial firms. Business angels are also far less regulated for minimum quality practices than their institutional equivalents. In Lerner’s arguments, there is an

implied “trickle down” process with classic venture capitalists first reviewing prospects and the huge majority they reject becoming part of the raw material of informal investors. Accordingly, he argues that we have at present little proof that fiscal incentive programs funded by the state are necessary to increase business angel activity, nor do we have a clear understanding as to how such programs may best be structured in order to realize policy goals (Lerner, 2002). While Lerner’s argument remains cogent, the assumption that business angels are second best to institutional venture capitalists when appraised by early-stage investment performance remains hotly debated.

Lerner is supporting those academics who argue that there exists a failure in quality demand (“investment readiness” argument) rather than a lack of available risk capital for nascent enterprises (the “equity gap” argument). Yet, in practice, the minimum size thresholds for new investments imposed by the majority of professional venture capitalists are now so high that it is highly unlikely that they would consider investing in a seed or start-up investment other than for the most interesting opportunity in a new technology. Such speculative and exploratory investment could be viewed as placing a “put” option on potentially important future developments (McGrath and McMillan, 2000; Miller and Arikan, 2004). Yet, as Dimov and Murray (2006) demonstrate, such an integrated, market intelligence strategy is rarely undertaken by professional venture capital investors outside the largest US firms.

Business angels – future policy interest

Business angels by their very scale and ubiquity remain a focus of considerable interest to governments. Ironically, it is their lack of activity that makes them so enticing. A modest increase in informal investment activity is likely to have a much larger impact than an equivalent increase in institutional venture capital given the business angels’ predominant focus on early-stage investment. Accordingly, it is likely that business angels will increasingly feature in the enterprise policy activities of developed economies. Countries that facilitate and incentives personal investment activity via generous fiscal incentives are likely to have an important head start. A number of trends will re-inforce the growing policy interest in business angels. As the maturing, venture capital industry gets larger in

terms of funds managed (rather than the numbers of venture capital general partnerships), a number of traditional venture capital funds will continue to withdraw from the earliest stage activities as partners experience the need to invest large amounts of fund monies quickly. The relative performance advantages of later stage funds, including private equity investments, will exacerbate this long term trend in Europe. In order to support this rational trend by their venture capital firm members, national venture capital industry associations will increasingly, work with government to ensure that business angels receive public support.

Such a co-operative industry stance supports the supply of potentially attractive businesses to their venture capital firm members (at later rounds of finance) as well as deflecting government concern of the limited impact of institutional venture capital on national innovation and entrepreneurship programs. Business Angel Networks will be able to exploit this opportunity to argue successfully for greater support (operating grants, co-investment schemes, etc.) for individual investors and (increasingly) legally defined angel groups. Programs supporting networks development, information exchange and investor training, are likely to continue to attract public support at regional, national and international levels. Yet, few public supported programs are likely to invite independent academic evaluation and scrutiny *de novo*. Hence the importance of Sohl's call for more support for empirical research into a major financing activity that, in comparison to its institutional venture capital equivalent, can still reasonably be viewed as *terra nullis*.

WHAT HAVE WE LEARNED FROM PUBLIC INVOLVEMENT IN PRIVATE VENTURE CAPITAL ACTIVITY?

One can argue with some authority that the evidence of government learning in the arena of early-stage risk capital finance is rather poor. Detractors of government's role could repeatedly point to unviable small fund sizes in public supported programs; the willingness of governments still to manage their own business angel programs outside the incentives and disciplines of the market; the imperfect integration between governments' entrepreneurial agendas, their fiscal programs and investors' interests³¹; the still under-developed role of informal investors; and the poor financial returns on public supported

funds. The vigorous growth over the last quarter of a century when venture capital has grown to become a major asset class is not reflected in equal successes for public involvement in new enterprise financing.

However, the choices which government face are no often easy or unambiguous. Governments with responsibilities for departmental portfolios of often competing interests have to determine if they wish to intervene in order to correct market imperfections or to realign incentives in a manner congruent with their policy goals. They have to decide how attractive a flourishing venture capital industry would be to the domestic economy, and what can sensibly be done to promote its realization. Yet, while there is considerable and growing governmental activity in the arena of early-stage risk capital investment, the body of knowledge on which this activity is based remains sparse. Lerner (1999) has commented on the absence of theory in guiding public venture capital activity. Similarly, Jeng and Wells (2000) note that just as later and early-stage venture capital investment behaviors are different and not necessarily influenced by the same factors, so likewise are public and private venture capital activities different. These authors note that we are still relatively poorly informed as to the appropriate role of government in venture capital activity. We lack both a canon of theoretical understanding (and guidance) as well as a diversity of exemplar programs from which we can benchmark progress.

Gilson (2003) takes a similar view that governments often undertake programs without a clear understanding as to the full consequences of their actions. In addition, political cycles and investment cycles are rarely likely to be synchronized. There appears a growing academic consensus that sanctioning government intervention is a decision that should only be taken with considerable caution, and perhaps only when private venture capital markets are patently failing. Circumstantial evidence of the large number of sub-optimal, publicly supported venture capital programs operating across the world including the US would suggest that academics' concerns with the logic of public intervention has frequently been ignored by policy makers (Bazerman, 2005; Rynes and Shapiro, 2005).

Yet, to suggest little has been learned and acted upon would be too pessimistic a diagnosis. The ubiquity of entrepreneurial finance programs at local, regional, national

and, increasingly, international levels of government are now so omnipresent that some learning is inevitable. There is a burgeoning corpus of research knowledge from scholars based in the entrepreneurship, innovation, management and economics subject areas³². However, the communication and accommodation of research lessons into contemporary policy actions at national level is still capable of considerable improvement (Söderblom and Murray, 2006).

Further, and of equal importance, government usually works in the least attractive areas of a financial market. The public exchequer becomes involved because of the absence of private investors. Accordingly, the situation of difficult investment choices and poor returns is virtually guaranteed. Governments should not be criticized for poor returns *per se*. Rather, a more apposite question is whether they should rationally hazard public monies by attempting to undertake investment activities in areas so commonly abandoned by informed, experienced and profit-seeking commercial interests.

These comments should not be seen as an *apologia* for harassed policy makers. There is regrettably little evidence that robust research findings are acted upon in new policy formation and execution. It is inevitable that partially informed program designs will have real (and avoidable) costs. Sometimes, in the absence of institutional venture capital research programs, government may too readily accept the convenient results of management consultants or venture capital industry association³³.

To summarize the factors that policy makers may consider in their efforts, a set of broad guidelines based on our imperfect contemporary knowledge is offered:

- Institutional or informal venture capital programs should harness private investors' interests and experience as agents of government program goals. Any deviation from this norm should be rigorously evaluated prior to agreement.
- There are trade-offs between venture capital or business angel program outcomes. Policy makers should be explicit as to the 'value' of different objectives, e.g. return on capital, innovation, employment, regional investment etc. in both launching and evaluating programs.

- There are major economies of scale and scope in venture capital fund activities. Program outcomes should be modeled and simulations tested prior to committing public funds to unviable fund structures.
- The levels of unmanageable uncertainty at the very earliest stages of venture capital investment in novel technologies may be such that it may not be sensible to allocate resources by markets alone. Venture capital should be seen as only one of a range of financing options that may also include merit-based grants and other support.
- The premium for managerial and investor experience is high in informed, professional labor markets. Program designers need to reflect on how such tacit knowledge may be best harnessed to address the challenges of early-stage investment.
- The USA has provided venture capital communities world-wide with a huge stock of experience and expertise. Much of this learning may be valid and relevant outside North America. Some of it will not be usable in other and different national contexts. It is implausible that a globalizing venture capital industry will, over time, be reliant on one absolute and exclusive model of success.
- All new venture capital programs involving public funds should have a formal (and independently validated) evaluation model built into the program design stage.

FUTURE ACADEMIC RESEARCH OPPORTUNITIES

Venture capital studies have reached their adolescence – a period of rapid but awkward growth. In a subject area originally colonized by entrepreneurship and small business scholars, the field has burgeoned (in parallel with a wider interest in entrepreneurship since the early 1980s) to include other managerial disciplines and, above all, economics. While venture capital studies embraces the theories and methodologies of economics, finance, organizational behavior etc., policy studies by their very nature are pragmatic in purpose. Governments wish to know how to change or improve systems to achieve

tangible and cost effective outcomes – preferably quickly and by incremental and non-disruptive changes. Such utilitarianism still sits uncomfortably with some scholars holding purely theoretical interests in venture capital finance. This is not unreasonable. However, for others, the ability to research and influence the actual processes of government in an area with little established public expertise provides both intellectual and professional rewards.

With this latter group in mind, the following questions are presented as some domains in which our research knowledge is still wanting.

- Does the equity gap actually exist? If so, at what levels of finance, who is affected, and are the adverse consequences material?
- Measured by the criteria of venture capital fund survival and acceptable investment returns, are early-stage (seed and start-up) activities a long term viable proposition both within and outside the USA? What policy and operational conditions need to be in place to secure such desired commercial outcomes?
- Business angels are seen as an alternative to institutional venture capital providers at the earliest stages of investment. Is such an assumption empirically valid? By what means can business angels succeed in early-stage market conditions that are presently hostile to venture capital success?
- What actions under governments' control (e.g. tax incentives, equity enhancement, investor training, network support, etc.) are most effective in stimulating the investment activities of current and potential business angels?
- By what means can business angels and venture capital firms most effectively work together to support high potential young firms?
- By what means should public/private “hybrid” venture capital programs be evaluated in order to both capture the economic and social objectives of all participating investors and to allow meaningful cross-program comparisons?
- Venture capital has evolved to become one of a range of “alternative asset classes” by which financial institutions may seek to engineer the risk/reward profiles of their investment portfolios. The actors involved and the decision processes by which such institutional portfolios are designed remains a “black box”. How may researchers

address the dearth of empirical studies focusing on the nature of institutional investor decision making.

- As venture capital activity has internationalized so has the policy response. How may academics best respond collectively to international and comparative studies of venture capital activity?
- We now have an international and multi-disciplinary body of research into venture capital studies that has chronicled the introduction and growth of risk capital activity across a large number of developed economies in Europe, North America and beyond. Emerging economies in Asia, South America and Eastern Europe are showing considerable interest in the putative role of venture capital in supporting the genesis and growth of new enterprises and industries. How may academics feature in the processes by which emerging economies learn effectively from extant venture capital experience?

While academic scholars have much to offer their policy colleagues, it cannot be assumed that the potential complementarity of their interests will guarantee research access or funding. Scholars will have to earn policy makers' respect and active support. Experience shows that this is not an easy task. Similarly, while academics are often tribal in their disciplinary interests (see Sapienza and Vallaneuva's chapter), policy clients frequently prefer inter-disciplinary teams that will address big issues with strong evaluation and execution recommendations. The skill for the academic is to be able to meet legitimate executive needs while still being able to undertake studies capable of scholarly validation via peer review. Arriving at such mutually acceptable outcomes is not easy and will require a level of trust building and mutual understanding between academics and policy makers that is still largely in its infancy. Entrepreneurship and venture capital scholars are going to have to be equally as entrepreneurial in the crafting of venture capital policy research ideas as the creators and funders of the new enterprises on which their discipline is founded.

ENDNOTES

- ¹ Contemporary venture capital evaluations by government have included programs in Finland (2002), UK (2003), and Ireland (2005) as well as New Zealand in 2005. Only the Finnish and New Zealand evaluation is in the public domain (Maula and Murray, 2003; Lerner et al., 2005).
- ² We have tended to use the terms “knowledge-based firms” and “new technology-based firms” interchangeably. While this is a sensible short-hand in the context of this chapter, new technology-based firms should be seen as a specialist category of knowledge-based firms. They both share a reliance on tacit and intangible assets for their competitive advantage.
- ³ As Lerner (1996) has noted the US similarly has financed a considerable amount of public venture capital actions both at federal and state levels. This involvement continues to be material. ⁴Each of these economies has already experienced considerable inward investment by private venture capital and private equity firms. However, these commercial interests are rarely at the level of new enterprises.
- ⁵ See <http://www.indiaventurecapitala.org>.
- ⁶ BVCA statistics only record start-up deals and not seed investments. 208 companies received start-up investment in 2005.
- ⁷ Brazil, India and China.
- ⁸ In practice, early stage investors rarely seek interest payments but if successful are (ultimately) rewarded by a significant capital gain multiple at exit.
- ⁹ See Pickering (2002) for a valuable government policy maker/insider’s view of six UK reports on funding tech-based Small and Medium Sized Enterprises.
- ¹⁰ While senior policy makers are usually very aware of the limitations of transferring models to new contexts, this complexity does not stop politicians framing the question as noted.
- ¹¹ The statement is curious in that it implies a modest historic role for public policy in the US experience.
- ¹² If funds are ranked by the proportion of seed investments to total investments, a UK fund does not appear until position 59.
- ¹³ The Directory of Support Measures (EC 2003) lists seven measures by which the state can assist SMEs: Reception, facilities and basic information, referral; Professional information services; Advice and direct support; SME-specific training and education; Finance; Premises and environment; Strategic services.
- ¹⁴ There is some more encouraging evidence that early-stage investing in Europe post 2001 is showing more positive returns and that performance is not affected by location once fund structural characteristics are controlled (Lindstrom, 2006).
- ¹⁵ Over the last 25 years, Northern European countries have arguably had a stronger tradition of direct investment activity via public agencies than the more market driven Anglo Saxon models of the US and the UK.

- ¹⁶ Despite Rocha and Ghoshal's (2006) concern with the adversarial presumption of agency theory, it is a very accurate and powerful concept that has considerable salience to venture capital studies.
- ¹⁷ Over the period 1959 – 2002, this program was responsible for raising \$37.7 billion for some 90 000 businesses. (US Small Business Administration, 2002).
- ¹⁸ See the Small Business Services' notes for the proposed Enterprise Capital Fund http://www.sbs.gov.uk/SBS_Gov_files/finance/waterfall.pdf.
- ¹⁹ EVENTURE CAPITALA valuation guidelines exist to set a basis for performance comparison between funds.
- ²⁰ Communication with Yigal Erlich, the Government Chief Scientist of Israel at the time of the program's inception and now CEO of the Yozma Group, Tel Aviv. The seven emulating countries cited by Mr Erlich are: Australia, Czechoslovakia, Denmark, Korea, New Zealand, South Africa and Taiwan.
- ²¹ Finnish 2003 and Irish 2005 evaluations were on venture capital programs that, while involving state investment, could not easily be classified as hybrid funds using the term as understood by the OECD.
- ²² The public access of Finnish evaluations is an honorable exception to the rule.
- ²³ In this chapter no difference will be drawn between business angels and informal investors on the simplistic assumption that they both categories of private individual who provide risk capital (and loans) for non-family enterprises.
- ²⁴ In Europe in 2006, a private equity fund, Permira, launched an international fund of approximately €10 billion. At least four venture capital funds of over \$10 billion are scheduled for launch in 2007.
- ²⁵ University endowment programs and investment trusts for high net worth family dynasties have played an important early role in the development of the US venture capital industry (Bygrave and Timmons, 1992).
- ²⁶ See http://www.lse.co.uk/financeglossary.asp?searchTerm=&iArticleID=1646&definition=business_expansion_scheme.
- ²⁷ The Venture Capital Trust program which allowed retail investors to access venture capital funds and provided another source of capital for young businesses was similarly launched in April 1995 (see: <http://www.hmrc.gov.uk/guidance/vct.htm>).
- ²⁸ The Netherlands had a broadly similar tax incentive program for private investors starting in 1996 and known as the "Aunt Agaath scheme".
- ²⁹ Research suggests that male informal investors outnumber female investors by about 5:1.
- ³⁰ There is a dearth of large scale, quantitative 'matched sample' empirical studies whereby the outcomes of BA investment on recipient firms can be compared to the outcomes of alternative investment channels on comparable enterprises.
- ³¹ Government's low interest policies have helped fuel a property boom that has arguably been a direct substitute to informal investments in new enterprise. See for example, the contemporary UK and Irish economies.
- ³² See the international Norface program on Venture Capital Policy Research seminars (www.norface.org) instituted by Murray in 2005.

- ³³ Most national venture capital associations have research departments producing analyses of the benefits of venture capital activities albeit from a clearly articulated position of interest.

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Figure 5.1 Investment risks in new technology-based firms

- Exceptional technical entrepreneurs are rarely competent or experienced business managers
- Project assessment and due diligence are highly problematic in areas concerning 'leading edge' technologies
- Uncertainty is compounded by the need to analyze both technological feasibility and the existence of a sufficiently large and attractive market (often for a product which does not yet exist)
- The speed of the change and the threat of technological redundancy often require an extremely rapid rate of commercial exploitation
- Competitive response and the availability of alternative products/services is likely to be rapid in dynamic and attractive new technology markets
- Successful NTBFs need to grow, internationalise and develop second generation products in a very short time horizon. These imperatives place exceptional managerial, financial and technical demands on a new business
- The scarcity of large, liquid and technologically informed capital markets increases the uncertainty of the future financing of the investee firm and the profitable 'exit' of the venture capital investor.

Figure 5.2 Simulation model of the effect of fund size on management's and private partners' returns (Murray & Marriott, 1998)

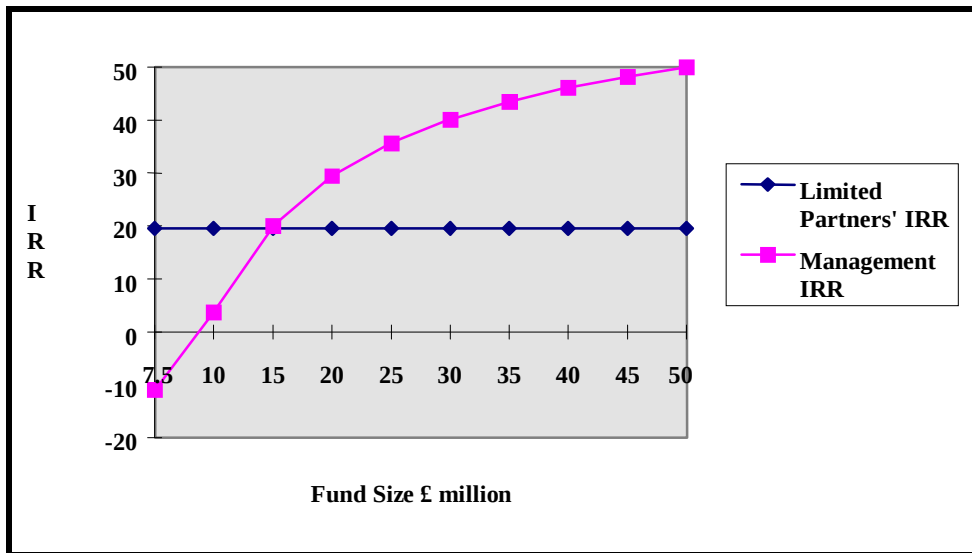


Figure 5.3 Negative consequences of small venture capital fund size

Small, early-stage venture capital funds bear disproportionately the following dis-economies:

- The high costs of reducing information asymmetries in immature, complex and dynamic markets
- The high levels of management support & guidance required by early-stage investees
- The limited ability to attenuate project risks by fully diversifying the venture capital fund
- The limited ability to invest large sums early in the life cycle of the investee firm
- The skewed risk/return profile resulting in the need for an exceptional success by the venture capital
- The long NTBF cycle and its implications on fixed term, fund structure/conduct/performance
- The limited ability to provide follow-on financing in a successful NTBF
- The danger of excessive dilution of ownership for the original investors in deal syndication

Figure 5.4 OECD venture capital policy recommendations

Box 1. VENTURE CAPITAL POLICY RECOMMENDATIONS	
Investment regulations:	
✦	Ease quantitative restrictions on institutional investors to diversify sources of venture funds
✦	Support the development of a private equity culture among institutional investment managers
✦	Facilitate creation of alternative investment pooling vehicles, such as funds-of-funds
✦	Improve accounting standards and performance benchmarks to reduce opacity of venture capital funds and protect investors
✦	Remove barriers to inflows of foreign venture capital finance
Taxation	
✦	Reduce complexity in tax treatment of capital from different sources and types of investments
✦	Decrease high capital gains tax rates and wealth taxes which can deter venture capital investments and entrepreneurs
✦	Evaluate targeted tax incentives for venture capital investment and consider phasing out those failing to meet a cost-benefit test
Equity programmes	
✦	Use public equity funds to leverage private financing
✦	Target public schemes to financing gaps, e.g. start-up firms
✦	Employ private managers for public and hybrid equity funds
✦	Consolidate regional and local equity funds or use alternative support schemes
✦	Focus venture funding on knowledge-based clusters of enterprises, universities, support services, etc.
✦	Evaluate public equity funds and phase-out when private venture market matures
Business angel networks	
✦	Link local and regional business angel networks to each other and to national initiatives
✦	Ensure linkages between business angel networks and technology incubators, public research spin-offs, etc.
✦	Provide complementary support services to enhance investment-readiness of small firms and increase demand
Second-tier stock markets	
✦	Encourage less fragmentation in secondary-stock markets through mergers, e.g. at Nordic or European level
✦	Enhance alternative exit routes such as mergers and acquisitions (M&As)

Figure 5.5 Five avenues to stimulate entrepreneurship

5. Demand side intervention

R&D expenditure, stimulating competition

6. Supply side intervention

Labor mobility, regional development

7. Influencing input factors

Higher education, venture capital market

8. Influencing preferences

attitude/mindset, image of entrepreneurship

9. Individual decision making process

Taxes, social security, level of benefits

Figure 5.6 UK government's model of building an enterprise economy (source: Small Business Service (2004a))

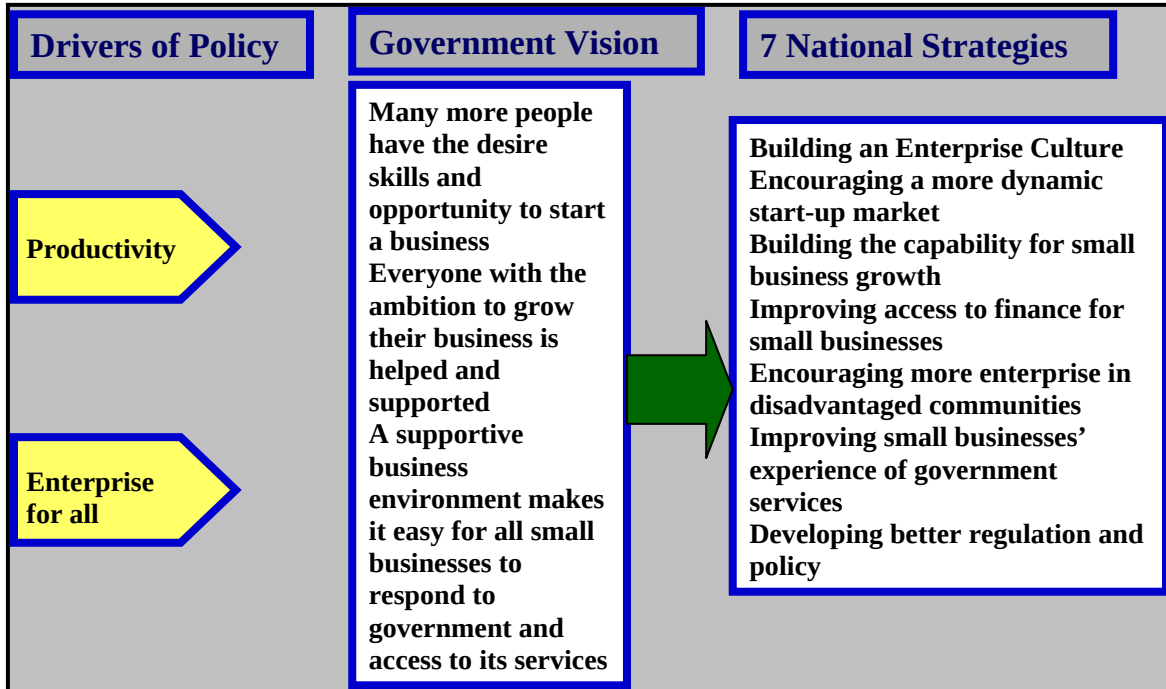


Figure 5.7 Generic 'hybrid' venture capital model (source: Small Business Service (2004b))

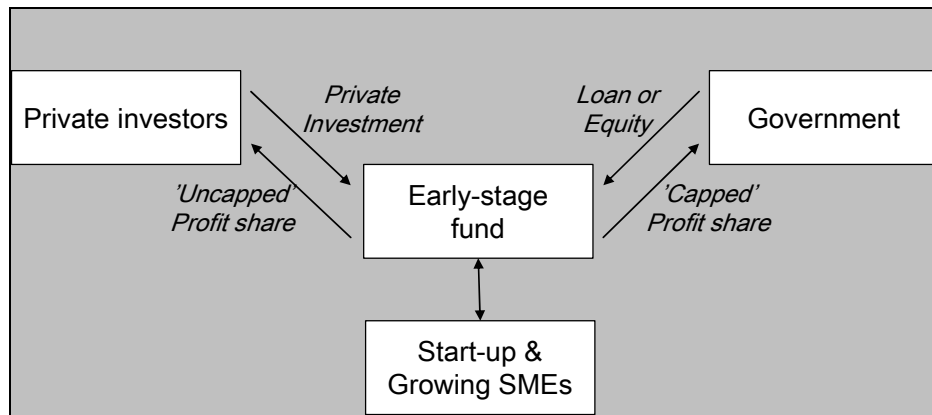


Figure 5.8 Enterprise investment scheme – investor tax benefits

- income tax relief - 20 % of amount invested in terms of tax - period to hold shares of 3 years
- exemption from capital gains tax on shares held for the 3 year period
- capital losses on shares treated as income losses
- deferral of chargeable gain on any asset
- maximum invested per tax year for income tax relief is now £400,000 (€589,000)

Table 5.1 Long run investment returns to venture capital and private equity in Europe

European Private Equity Funds Formed 1980-2005 Net Returns to Investors From Inception to 31 Dec 2005 THOMSON			
Stage	Pooled IRR	Upper Quartile	Top Quarter IRR *
Early Stage	0.1	2.3	13.6
Development	9.2	9.0	18.8
Balanced	8.3	8.5	23.7
All Venture	6.3	6.2	17.1
Buyouts	13.7	17.8	31.8
Generalist	8.6	8.8	10.3
All Private Equity	10.3	10.6	22.9

*The top quarter IRR is the pooled return of funds above the upper quartile

Updated Benchmarks 31-December-2005 – presented at EVCA Symposium 16 June 2006
Source: Thomson Financial (VentureXpert database)

evca
European Venture Capital Association

Table 5.2 Publicly financed venture capital “equity enhancement” schemes

Feature	Description	Incentive effects	Examples (present & past)
Public investor co-investing with private investors	Government matching the investments by private investors	Helps in setting up a fund. Also helps to build a sufficiently big fund to benefit from economies of scale. However, investing in <i>pari passu</i> with private investors does not have direct incentive effects i.e. it does not improve the expected returns for private investors in market failure segments such as early stage.	<50% of the fund: Europe/EIF Finland/FII Israel/Yozma >50% of the fund: Australia/IIF and Pre-seed Fund USA/SBIC and SSBIC UK/regional venture capital funds
Capped return for public investors	After the all the investors (including the public investor) have received certain IRR (e.g. interest rate + perhaps some risk premium) the rest of the cash flows are distributed to private investors only.	Capped return for the government increases the expected IRR for private investors. The incentive effect is such that it increases the compensation for good performance. This in turn creates a strong incentive for the private investors to incentivise the general partners to make successful investments and add value to portfolio companies	UK/regional venture capital funds Australia/IIF and Pre-seed fund Chile/CORFU
Buy-put option for private investors.	Private investors are given the option to buy the share of the government at (or until) a specific point of time at predetermined price (typically nominal price + interest rate)	The effect of buy-out option on the IRR of private investors is quite similar as the effect of ‘capped return’ of public investor. However, there are two additional benefits: 1) The buy-out option gives both the public investor and the private investors an opportunity to demonstrate success earlier and more visibly than in the capped return alternative 2) In the case of success, government gets a quick exit from the fund and can put the money to work again instead of waiting for the returns on fund termination	Israel / Yozma New Zealand / New Zealand Venture Investment Fund
Downside protection	Downside protection means the government	Downside protection has a negative effect from the incentive perspective. While	Germany / WFG Germany / tbG & KfW France / SOFARIS

	underwriting losses from the portfolio.	downside protection helps support the IRR, it decreases the difference in returns for private investors and the management company between successful and unsuccessful investments. The effects of good selection and value-added efforts have a lower impact on the performance of the fund.	Denmark/ Vaekstfonden (Loan Guarantee Scheme)
Fund operating costs	Government gives a subsidy for the management company to cover some of the costs from running the fund.	The fund operating costs are neutral from the perspective of incentives to fund management or LPs while increasing the IRR of the fund	Europe / European Seed Capital Scheme
Timing of cash flows	Ordering of the cash flows so that public investor puts the money in first and gets the money out last	The IRR of the private investor can be enhanced through timing of cash flows improving the attractiveness of the fund.	UK / Regional Venture capital Funds